



Does connectivity power solar energy? Evidence from the belt and road initiative

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ABSTRACT

Solar energy plays a critical role in the global energy transition but is constrained by economic and technological diffusion barriers. While the Belt and Road Initiative (BRI) is expected to mitigate these obstacles through international cooperation and trade facilitation, causal evidence of its impact on solar power deployment remains limited. This study treats the BRI as a quasi-natural experiment and employs a multi-period difference-in-differences model with panel data from 135 countries (2000–2022) to estimate its impact on solar power generation. The results indicate that: (1) Solar power generation in BRI-participating countries increased by about 39.8 % relative to non-participating countries on average. Considering the total growth along the BRI, this effect accounts for approximately 17.4 % of the solar power expansion. (2) Facility connectivity, financial integration and unimpeded trade are key channels. (3) Heterogeneity analysis reveals that the BRI's effect is more pronounced in countries with lower trade barriers, superior solar irradiation, and lower reliance on traditional energy sources. This study presents rigorous causal evidence on how international cooperation boosts solar energy development and provides policy recommendations to enhance solar and green energy collaboration among BRI countries. The findings hold significant implications for advancing a fair and equitable global energy transition.

1. Introduction

Currently, with climate change intensifying and extreme weather events becoming more frequent, the urgency for a global transition toward a low-carbon energy system has reached an unprecedented level [1–4]. Among various renewable sources, solar energy stands out for its vast untapped potential [5,6]. According to the International Energy Agency, newly installed PV capacity reached 452 GW in 2024, accounting for over 75 % of global renewable capacity additions, establishing solar energy as a key sustainable alternative [7]. However, the global distribution of solar resources and technologies remains uneven, and trade barriers have severely hindered the diffusion of solar energy. In this context, extensive international cooperation has become crucial for unlocking the full potential of solar development. The Belt and Road Initiative (BRI) has facilitated cross-border coordination in institutions, capital, and technology networks, thereby providing substantial momentum for global solar energy development.

Solar resources are abundant along the BRI, and the photovoltaic industry holds promising prospects for sustainable development [8]. However, many participating countries' dependence on fossil fuels remains high [9,10]. This reliance is not only due to the entrenched market position of fossil energy, but more importantly because solar development faces significant cost and technological barriers [11,12]. From installation to grid-scale integration, solar projects require substantial financial support. In addition, protectionist policies in various countries hinder the diffusion of solar technologies, constraining global energy transitions—a challenge particularly acute in developing economies [13–15]. As the world's largest exporter of photovoltaic products, China has leveraged the BRI to foster green energy partnerships by opening trade channels and providing investment in renewable energy projects [16,17]. Under this framework, new solar farms and PV plants have emerged across Central Asia [18]. The BRI power cooperation has become one of the key drivers of global renewable energy expansion, offering developing countries opportunities for cleaner and more

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efficient electricity generation [19–22].

However, empirical studies have yet to provide conclusive evidence on the effects of this major international cooperation under the BRI. Existing research has largely focused on the role of declining photovoltaic technology costs, efficiency improvements, and domestic policy support in driving solar deployment [23–25], while overlooking the critical role of international cooperation. Especially against the backdrop of the continuous escalation of the U.S.-China trade war in recent years, which has profoundly impacted the global solar supply chain [26], analysis of the actual effects of BRI cooperation on solar development remains absent. Based on international panel data from 135 countries (2000–2022), this paper establishes a multi-period DID model. Using the BRI initiative as a natural experiment, it examines the impact of BRI energy cooperation on solar power generation and thoroughly investigates the theoretical mechanisms behind this influence. Furthermore, it conducts heterogeneity analysis from the perspectives of trade barriers and resource endowment. This study holds significant implications for understanding policy outcomes and deepening international energy cooperation.

This paper makes three primary contributions: (1) The energy transition effects of the BRI in the solar sector have not been comprehensively quantified. This paper employs a multi-period DID model to tackle the potential endogeneity issues in the policy and empirically analyzes the impact of the BRI on solar energy development in participating countries, aiming to explore the specific role of this international cooperation platform in advancing SDG 7. The results show that the BRI increased solar power generation, accounting for approximately 17.4 % of the observed growth. The study provides causal evidence from the perspective of energy end-output for evaluating the effectiveness of BRI solar cooperation. (2) This study examines the mechanisms through which the BRI influences solar energy development across three key dimensions—infrastructure, investment, and trade. While prior research has established BRI's role in mitigating trade sanctions, enhancing network resilience, and boosting photovoltaic exports [27,28], the initiative's broader impact on renewable energy extends beyond trade, given its scale and transcontinental reach connecting Asia, Africa, and Europe [29]. Grounded in the policy vision of BRI, this mechanistic evidence advances the theoretical understanding of how international cooperation platforms catalyze renewable energy transitions. (3) Provide policy recommendations for governments to optimize international cooperation, offering valuable insights and references for advancing global renewable energy development. By conducting heterogeneity analysis from the perspectives of trade barriers and resource endowment, this research incorporates critical institutional and environmental factors in constructing a multilateral energy governance system. The study reveals that the BRI's impact is more pronounced in regions with lower trade barriers, superior solar irradiation conditions, and lower dependence on traditional energy sources, thereby supplying a scientific foundation for formulating targeted policies.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature and elaborates on the theoretical mechanisms. Section 3 develops the empirical econometric model and details the selection of variables and indicators. Section 4 presents the empirical results and discusses the key findings. Finally, Section 5 concludes the paper and puts forward targeted policy recommendations.

2. Theoretical analysis

2.1. Solar energy and the Belt and Road Initiative

Among the renewable-energy sectors championed by the BRI, solar power stands out for its breakneck growth and vast potential, making it pivotal to the global net-zero agenda [30]. Even under reasonable assumptions about land use and conversion efficiency, the recoverable potential of solar power remains significantly higher than that of other renewable and non-renewable sources—estimated to meet global

primary energy demand 12 times over each year [6]. Research has largely highlighted two engines behind its rapid scale-up: the sharp fall in PV costs and steady gains in conversion efficiency [24,31]. At the same time, the role of policy support mechanisms in driving solar deployment has been well documented. Feed-in tariff subsidies, tax credits and similar measures have been shown to significantly stimulate market growth [23,25,32]. These studies reveal the general patterns behind solar expansion.

China's green energy strides offer invaluable lessons for global environmental governance [33]. The 2013-launched BRI rapidly expanded China's global footprint, explicitly prioritizing "green energy cooperation" as a key pillar [34]. Against the backdrop of the BRI, existing research has been more focused on photovoltaic trade. Zhu et al. [28] using Chinese customs trade data from 2009 to 2016, quantitatively confirmed that BRI markets helped Chinese PV firms mitigate losses from trade sanctions. Hu et al. [27] assessed the robustness of the BRI PV trade network, highlighting the critical role of the "Five Connectivity" components. Qian et al. [35] and Sun et al. [36] have explored site selection for PV development along the BRI. These studies focus on trade and industrial dimensions, offering references for understanding the internationalization path of the BRI photovoltaic industry.

Broader studies quantify the policy effects of the BRI from the overall perspective of energy cooperation. Power infrastructure projects have been one of the key areas of interest for Chinese enterprises and investors under the BRI framework [37]. Scholars have focused on how the BRI affects energy efficiency in participating countries, with findings suggesting that the initiative has a particularly significant impact in countries within the middle-to-upper efficiency quantiles [38]. Economic growth and shifts in industrial structure have been identified as key indirect pathways through which the BRI exerts its influence [39]. Other studies have shown that the BRI has helped raise electrification levels in participating countries by around 1 % [40]. Xu [41] found a strong positive correlation between BRI implementation and the growth of renewable energy. Chishti et al. [42] also affirm the BRI's contribution to the global energy transition. In addition, several researchers have affirmed the BRI's role in reducing carbon dioxide emissions [43,44]. Most of these studies focus on the energy system as a whole, with limited attention given to solar energy—a key subsector within the broader field of renewable energy.

In summary, the existing literature exhibits three key limitations. First, while global-level studies have identified general patterns in solar energy development, they fail to account for the distinctive policy and market context of the BRI. Second, much of the BRI-related research centers on photovoltaic trade and industrial chains, with limited attention given to its actual impact on power generation outcomes. Third, although energy economists have conducted in-depth analyses of BRI-driven energy cooperation, these studies predominantly operate at the aggregate energy system level, lacking granular examinations of specific subfields. Solar energy plays an increasingly pivotal role in the global energy transition, particularly in addressing energy poverty and facilitating low-carbon development in less developed countries. In light of these gaps, this study employs a multi-period DID model to systematically assess the impact of the BRI on solar power generation in participating countries, thereby contributing empirical evidence and policy insights into the initiative's role within targeted energy subsectors.

2.2. The mechanism of facility connectivity, unimpeded trade and financial integration

Under the BRI, China has promoted the "Five Connectivity" framework—policy coordination, facility connectivity, unimpeded trade, financial integration, and people-to-people ties—gradually transforming BRI into a global platform for development cooperation. The BRI has increasingly served as a platform for knowledge sharing, particularly in renewable energy sectors like solar power. While all five priorities reinforce each other, three prove especially pivotal for solar

collaboration: facility connectivity, unimpeded trade, and financial integration.

With regard to facility connectivity, the BRI has built physical foundations for solar development. Because many BRI participants are developing countries, distributed solar has become a key tool for easing energy poverty and advancing low-carbon transitions. However, structural hurdles still hold back large-scale and sustainable deployment [45]. Ageing distribution grids struggle to absorb high shares of distributed generation. The BRI prioritizes infrastructure connectivity, fostering cross-border collaboration and development in critical sectors such as power grids, transportation networks, and energy infrastructure. Chinese firms have built PV power stations in numerous BRI countries, boosting local solar installed capacity [46]. For instance, in Pakistan, the China-built solar plant boosted the country's grid-connected PV capacity through supporting transmission infrastructure [47]. This expanded infrastructure directly enables higher electricity generation.

In terms of unimpeded trade, the BRI has accelerated solar product flows. Through tariff reductions, trade agreements, and logistics streamlining, BRI has energized solar product trade between China and partner economies [48]. The launch of dedicated China-Europe Railway Express routes for PV products—such as the Xi'an–Rotterdam line—which significantly shortens delivery times, serves as a strong example of enhanced trade facilitation under the BRI framework. With China's PV exports growing steadily, BRI regions now rank among its top export destinations [49]. Enhanced cross-border movement of equipment and technology actively supports solar generation growth in these markets [27,28].

With respect to financial integration, the BRI has fueled solar investment. Unlike operational phases, solar technology demands substantial upfront capital [11,50]. However, in many developing regions, financial systems provide inadequate support for renewable energy initiatives. Persistent challenges such as elevated financing costs, weak credit frameworks, and extended project payback periods significantly constrain the viability of solar energy projects [51,52]. Since 2015, China's green finance policies have explicitly favored renewable projects [3]. Central investment institutions have increased portfolio allocations to non-hydro renewables, cementing China's role as the leading public financier for power sectors in developing economies [53–55]. By unlocking project financing, BRI bridges critical electricity gaps and amplifies solar generation capacity.

3. Research design

3.1. Econometric model

Treating the BRI as a quasi-natural experiment, this study examines how the policy affects solar power generation in participating countries. A multi-period DID model is constructed as follows:

$$\ln solar_{it} = \alpha_0 + \beta_1 DID_{it} + \gamma Control_{it} + \alpha_t + \lambda_i + \varepsilon_{it} \quad (1)$$

where $\ln solar_{it}$ represent the logarithm of solar power generation for country i in year t . The key explanatory variable is defined as: $DID_{it} = Treat_i \times Post_t$. $Treat_i$ is a binary indicator equal to 1 if country i has joined the BRI and 0 otherwise. $Post_t$ is a time dummy that equals 1 for the year of signing the Memorandum of Understanding (MoU) and all subsequent years, and 0 for earlier years. For countries that signed the MoU in December, we assume they did not receive any practical influence from the BRI during that calendar year; therefore, $Post_t = 0$ for the signing year in such cases. The model also includes country fixed effects λ_i and year fixed effects α_t to control for common global shocks and time-invariant heterogeneity, respectively. The error term is denoted by ε_{it} .

3.2. Variable selection

3.2.1. Explained variable

The explained variable in this study is solar power generation. This indicator captures key information such as the actual operating efficiency of solar facilities, grid-connected utilization, and annual operating hours. Compared with installed capacity, power generation provides a more direct measure of the terminal value delivered to the electricity system, and thus more accurately reflects the effectiveness of energy policies and their macro-level outcomes. Solar power generation has been extensively applied in research on energy policies and the development of renewable energy [56,57].

3.2.2. Core explanatory variables

Following prior studies [58,59], we define BRI participation as the year a country signed the MoU. Countries that signed the MoU are classified as the treatment group, while non-signatory countries serve as the control group. To ensure robustness, we also conduct a sensitivity check using the earliest appearance of the terms "Belt and Road Initiative" and "solar energy" in the media as an alternative definition of BRI engagement.

3.2.3. Mechanisms

Based on the theoretical framework outlined in Chapter 2, the following mechanism variables are selected: (1) Facility connectivity (FC): FC is evaluated from both the capacity expansion and utilization efficiency dimensions. Capacity expansion is measured by one-year lag of each country's installed solar capacity, which reflects the physical infrastructure supporting solar power generation while mitigating simultaneity bias. We also include the share of grid-connected solar power in total solar power generation to reflect the efficiency of capacity utilization and grid integration. (2) Unimpeded Trade (UT): Assessed from both trade scale and trade facilitation perspectives. The first indicator is China's exports of PV products to BRI countries; this variable reflects the trade channel through which solar equipment and components are made more accessible, directly reducing deployment costs. The second indicator is the World Bank Logistics Performance Index (LPI), which further capture trade facilitation and border efficiency beyond pure trade volume. (3) Financial integration (FI): Captured by the volume of China's outbound energy investment, assessing the financial dimension of BRI cooperation in the energy sector.

3.3. Sample selection and data sources

This study constructs a panel dataset covering 135 countries from 2000 to 2022, including 107 BRI countries and 28 non-BRI countries—a distribution that closely mirrors the global proportion of BRI signatories. The explained variable, solar power generation, is sourced from the International Renewable Energy Agency (IRENA), which is a widely recognized and authoritative data provider [60,61]. Data for two indicators of FC also comes from IRENA. The timing of BRI participation is based on MoU signing years, drawn from Green Finance & Development Center reports.

Data on FI follows the approach of Lin and Bega [62], including energy-sector investments from the China Global Investment Tracker. To expand coverage, we also combine records from the China Overseas Finance Inventory and the China Global Energy Finance Database. In countries where investment data is absent, values are coded as zero, assuming no significant financing activity. UT is captured using data from the UN Comtrade database and the World Bank. Specifically, we extract trade flows under HS code 854140, which includes key PV products [63]. Descriptive statistics and variable construction methods are summarized in Table 1. All control variables—from GDP to CWD—are sourced from the World Bank.

Table 1
Descriptive statistics.

Variables	Description	Obs	Mean	Std. Dev.	Min	Max
<i>Insolar</i>	ln(Solar power generation output)	3294	2.144	3.532	−6.908	12.967
<i>DID</i>	Policy shock	7378	0.111	0.314	0	1
<i>GDP</i>	ln(Gross domestic product per capita)	6995	8.351	1.647	3.133	12.455
<i>Labor</i>	ln(Labor force)	6354	14.834	1.888	9.73	20.476
<i>FA</i>	Fixed asset investment/GDP	5564	23.652	8.481	−15.917	76.782
<i>Energy_resource</i>	Fuel, ores and metals exports/merchandise exports	4856	23.504	27.985	0	99.961
<i>IFDI</i>	ln(Inward foreign direct investment)	4616	9.299	2.483	−4.605	16.372
<i>Trade</i>	Exports of goods and services/GDP	5848	40.07	30.363	0.005	433.836
<i>SD</i>	Summer days in a year	7242	222.607	135.282	0	365
<i>CWD</i>	Consecutive wet days in a year	7242	41.563	46.111	0.19	261.09
<i>FC_Lag</i>	Facility connectivity, proxied by L.ln(solar installed capacity)	3259	1.973	3.575	−6.908	12.88
<i>FC_Grid</i>	Facility connectivity, proxied by grid-connected solar generation/total solar generation	1267	0.683	0.341	0	1
<i>FI</i>	Financial integration, proxied by ln(China's outward energy investment+1)	7378	0.009	0.047	0	1.125
<i>UT_Trade</i>	Unimpeded trade, proxied by ln(China's export of PV products)	3066	13.991	3.82	1.099	22.844
<i>UT_LPI</i>	Unimpeded trade, proxied by Logistics Performance Index	1071	2.868	0.587	1.21	4.3

4. Empirical results

4.1. Benchmark model estimation results

Variance Inflation Factor (VIF) values were calculated for all regressors, and none exceeded the threshold of 10, indicating that multicollinearity is not a significant concern in our model. Table 2 presents the results of Model (1), which investigates the impact of the BRI on solar power generation across countries. Columns (1)–(3) progressively introduce fixed effects. In all specifications, the BRI coefficient is positive and statistically significant at the 1 % level, indicating that participation in the BRI is associated with a substantial increase in solar electricity output. Economic development and labor force size are key positive drivers of solar power generation growth, whereas *FA* consistently exhibits a significant inhibiting effect. This suggests that, on average, domestic investment has not been effectively channeled into the solar sector. *Energy_resource* reflects a country's reliance on conventional energy and shows a mild constraining effect on solar development. In addition, the effects of *Trade* and *IFDI* are not uniform but are instead moderated by country-specific and time-specific factors. Finally,

Table 2
The impact of BRI on solar power generation.

	(1)	(2)	(3)
	Insolar	Insolar	Insolar
<i>DID</i>	3.115*** (28.78)	1.513*** (12.59)	0.335*** (2.91)
<i>GDP</i>	0.741*** (9.14)	1.910*** (9.33)	0.718*** (3.93)
<i>Labor</i>	0.452*** (6.68)	6.747*** (10.56)	0.968** (2.12)
<i>FA</i>	−0.014* (−1.82)	−0.080*** (−6.35)	−0.032*** (−3.71)
<i>Energy_resource</i>	−0.021*** (−10.61)	−0.009 (−1.52)	0.003 (0.68)
<i>Trade</i>	−0.012*** (−5.27)	0.022*** (3.45)	0.008 (1.46)
<i>IFDI</i>	0.639*** (8.97)	0.512*** (3.98)	−0.615*** (−6.14)
<i>SD</i>	0.004*** (6.50)	0.194*** (11.87)	0.002 (0.18)
<i>CWD</i>	−0.007*** (−6.11)	0.073*** (4.05)	0.024 (1.63)
<i>Constant</i>	−17.104*** (−15.54)	−162.938*** (−17.79)	−12.978* (−1.75)
<i>Country FE</i>	NO	YES	YES
<i>Year FE</i>	NO	NO	YES
<i>Observations</i>	2062	2060	2060
<i>R-squared</i>	0.534	0.790	0.878

Notes: Robust t-statistics are shown in parentheses. *, **, *** indicate significant at 10 %, 5 %, 1 % confidence levels respectively. The same as below tables.

the impact of interannual climate fluctuations on solar output is negligible compared to long-run national characteristics and macro-year trends.

We take Column (3) as preferred specification. The estimated coefficient of 0.335 implies that, on average, countries joining the BRI experienced an approximately 39.8 % increase in solar power generation relative to non-participating countries. To further assess BRI's importance for the overall trend of solar power generation, we follow the approach of Fan et al. [64]. Between the year of BRI accession and 2022, the average solar power generation among BRI countries increased from 652.672 GWh to 4493.647 GWh, corresponding to a logarithmic growth of 1.930 log points ($\ln(4493.647) - \ln(652.672) = 1.930$). Given that the average treatment effect of the BRI is 0.335, we estimate that the initiative accounts for approximately 17.4 % ($0.335/1.930 \approx 0.174$) of the total increase in solar power generation across participating countries. This is a substantial share, especially considering that national solar development is influenced by a wide range of other factors, including domestic policy environments, market dynamics, and technological adoption.

4.2. Robust test

4.2.1. Parallel trend test

In applied DID research, satisfying the parallel trends assumption is a critical prerequisite. It requires that without the policy intervention, treatment and control groups would follow identical trends [65]. Since we cannot alter policy implementation, scholars typically use event study methods to test pre-treatment trends [66]. If there is no significant difference in the pre-treatment trends, it can to some extent prove that the trends without policy intervention are the same. Fig. 1 shows that the pre-treatment coefficients are statistically insignificant, satisfying the parallel trends assumption. The promoting effect is strongest in the early and middle stages after joining. For early joiners, solar energy projects under the BRI framework are largely completed or nearing saturation by the sixth year, reducing new project development and weakening growth momentum.

4.2.2. Placebo test

This study conducted placebo tests to mitigate confounding factors. By systematically randomly selecting samples as fictitious treatment groups and repeating this process 500 times, the distribution of fictitious regression coefficients is shown in Fig. 2. The placebo estimates follow an approximately normal distribution centered around zero, while the actual estimated coefficient—marked by the solid line—lies well outside this range. This significant deviation between the true treatment effect and the placebo distribution supports the robustness of the baseline results and suggests that the observed impact is unlikely to be driven by random chance or omitted variables.

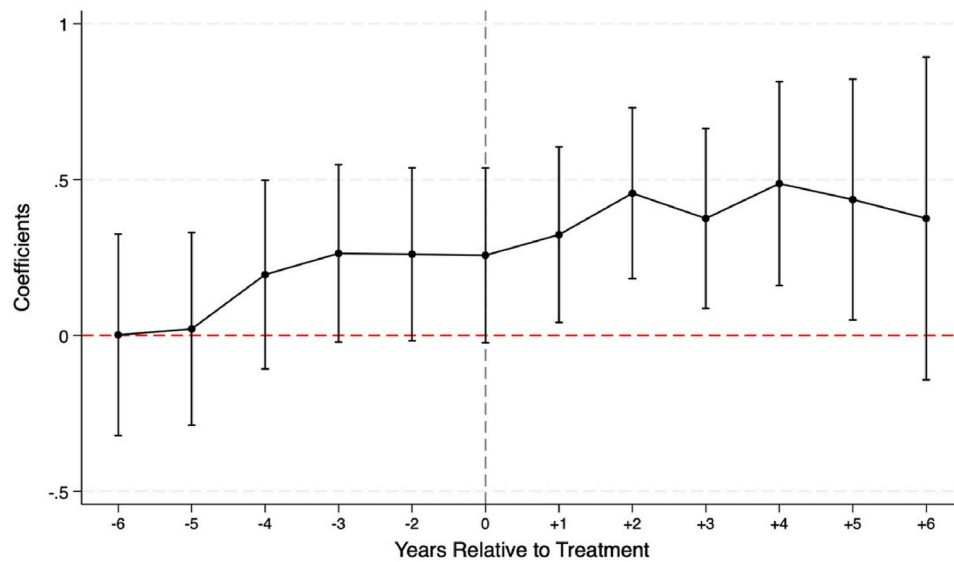


Fig. 1. Event study method.

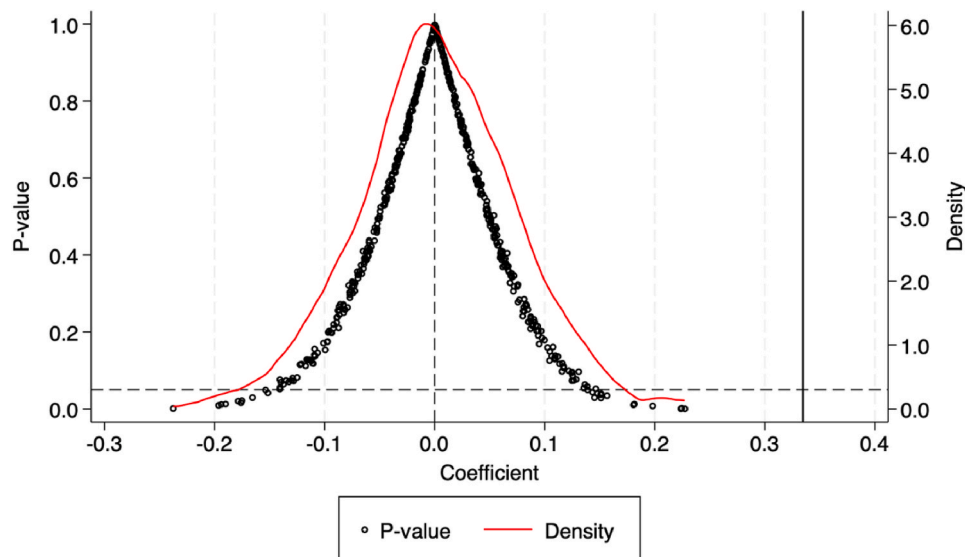


Fig. 2. Placebo test.

4.2.3. PSM-DID

Considering that developing countries may be more proactive in joining BRI due to infrastructure gaps and energy shortages, this study addresses such sample selection bias by re-estimating the baseline model using the PSM-DID approach. First, all control variables were selected as matching variables, with samples matched using kernel matching and 1:1 nearest-neighbor matching techniques. Fig.A1 and Fig.A2 in Appendix A display standardized bias for each variable, confirming successful balance tests. After matching, the DID method was reapplied to matched samples. As reported in Table 3, the treatment effect remains statistically significant and comparable in magnitude to the baseline estimates. This further reinforces the conclusion that the BRI has a robust and positive impact on solar power development in participating countries.

4.2.4. Omitted variable test

Although our baseline estimations already control for country-level characteristics and time effects, one may question whether domestic solar-energy policies could interfere with our results. To mitigate

Table 3
PSM-DID estimation results.

	Kernel matching	Nearest neighbor matching
	Insolar	Insolar
<i>DID</i>	0.262** (2.26)	0.270** (2.33)
<i>Constant</i>	-10.161 (-1.33)	-10.512 (-1.38)
<i>Control</i>	YES	YES
<i>Country FE</i>	YES	YES
<i>Year FE</i>	YES	YES
<i>Observations</i>	2007	2003
<i>R-squared</i>	0.875	0.875

potential endogeneity arising from omitted variables, we additionally control for the following policy variables: (i) Feed-in Policies (FIPs). Following Xu and Yang [67], years with feed-in subsidies are coded as 1 and those without as 0. Data are obtained from REN21, *Renewables 2023 Global Status Report*. (ii) Carbon tax policy. As a key form of carbon

pricing, the carbon tax variable is coded as 1 for years in which a carbon tax is implemented (and thereafter), and 0 otherwise. Data are sourced from the World Bank's Carbon Pricing Dashboard. (iii) Environmental tax. This variable reflects a country's environmental tax burden and serves as a form of negative incentive. Data are collected from the OECD Data Explorer. The results, as shown in Table 4, indicate that the BRI's positive effect on solar power generation remains consistently significant whether the policy variables are added individually or jointly. Environmental tax is found to promote solar development, whereas FIPs and carbon tax do not exhibit statistically significant impacts.

4.2.5. Refining the timing of policy intervention

While most international panel studies use MoU signing dates as cutoffs, some scholars adopt 2013—the year the BRI was officially announced—as the policy node [68,69]. We therefore reset the policy timing to 2013. As shown in Column (1) of Table 5, the estimated effect of BRI participation on solar power generation remains positive and statistically significant.

Next, given BRI's extensive energy collaborations, we substituted policy shock timing with the earliest news appearance of both "Belt and Road" and "solar energy" per country. This approach reflects actual public or institutional awareness of the BRI's relevance to solar development. We source this data primarily from the Factiva news database, a widely used platform for media monitoring and policy tracking. For countries with limited representation in Factiva, we complement our search with Chinese-language keywords and local media archives. The results based on this alternative timing, reported in Column (2) of Table 5, are also statistically significant.

Furthermore, for countries with recorded solar power generation but no media coverage linking BRI to solar energy, we assume that they were effectively untreated, not meaningfully influenced by the BRI in this sector. We re-estimate the model after recoding these countries as part of the control group. As shown in Column (3), BRI's solar-boosting effect remains significant.

4.2.6. Policy uniqueness test

During the study period, solar power generation in BRI countries may have also been influenced by concurrent policy initiatives. In 2012, China and 16 Central and Eastern European countries launched the "16 + 1 Cooperation" framework—a multilateral platform focused on infrastructure, trade, and investment—exhibiting significant overlap with BRI [70]. Importantly, these countries appear in both treatment and control groups, creating interference that cannot be isolated by DID methods. To address this potential confounding, this study excludes these 16 countries from the sample and re-estimates the model. As shown in Column (4) of Table 5, the estimated effect of the BRI remains positive and statistically significant after eliminating concurrent policy

Table 4
Inclusion of additional control variables.

	(1)	(2)	(3)	(4)
	Insolar	Insolar	Insolar	Insolar
<i>DID</i>	0.319*** (2.76)	0.835*** (2.60)	0.407*** (3.04)	0.919*** (2.82)
<i>FIPs</i>	0.221 (1.64)			−0.128 (−0.37)
<i>Carbon tax</i>		0.415 (1.46)		0.376 (1.30)
<i>Environmental tax</i>			0.324** (2.53)	0.534** (2.35)
<i>Constant</i>	−13.062* (−1.76)	6.342 (0.27)	−1.182 (−0.12)	11.197 (0.41)
<i>Control</i>	YES	YES	YES	YES
<i>Country FE</i>	YES	YES	YES	YES
<i>Year FE</i>	YES	YES	YES	YES
<i>Observations</i>	2060	451	1609	447
<i>R-squared</i>	0.878	0.871	0.882	0.870

Table 5
Other robustness tests.

	(1)	(2)	(3)	(4)	(5)
	Insolar	Insolar	Insolar	Insolar	Insolar
<i>DID₂₀₁₃</i>	0.790*** (5.75)				
<i>DID_{news}</i>		0.552*** (4.72)			
<i>DID_{new}</i>			0.335*** (2.91)		
<i>DID</i>				0.324*** (2.60)	0.701*** (3.34)
<i>L_{Insolar}</i>					0.616*** (15.75)
<i>AR(1)</i>					0.000
<i>AR(2)</i>					0.578
<i>Hansen</i>					0.288
<i>Control</i>	YES	YES	YES	YES	YES
<i>Country FE</i>	YES	YES	YES	YES	YES
<i>Year FE</i>	YES	YES	YES	YES	YES
<i>Observations</i>	2060	2060	2060	1852	1780
<i>R-squared</i>	0.879	0.879	0.878	0.885	

influences; BRI continues to exert a significant positive effect on solar power generation along the route.

4.2.7. Dynamic panel bias

To address potential bias in fixed-effects estimates arising from the dynamic persistence of solar power generation, this study conducts a robustness check using the System GMM approach. This method is particularly suited for panel data with potential endogeneity and autocorrelation concerns in the dependent variable. As shown in Column (5) of Table 5, the results from the System GMM estimation confirm the reliability of our findings. The model detects first-order autocorrelation but finds no evidence of second-order autocorrelation, satisfying the Arellano-Bond assumption for valid instruments. Additionally, the Hansen test for overidentifying restrictions fails to reject the null hypothesis, indicating that the instruments used are valid and exogenous. Importantly, the coefficient on the BRI treatment remains positive and statistically significant, reinforcing the conclusion that BRI has a robust and causal effect on solar power development in participating countries.

4.3. Influence mechanism analysis

Based on the theoretical framework laid out in Chapter 2, this paper investigates the mechanisms through which the BRI promotes solar energy development, focusing on three key channels: facility connectivity (Formula 2), unimpeded trade (Formula 3) and financial integration (Formula 4).

$$FC_{it} = \alpha_0 + \beta_1 DID_{it} + \gamma Control_{it} + \alpha_t + \lambda_i + \varepsilon_{it} \quad (2)$$

$$UT_{it} = \alpha_0 + \beta_1 DID_{it} + \gamma Control_{it} + \alpha_t + \lambda_i + \varepsilon_{it} \quad (3)$$

$$FI_{it} = \alpha_0 + \beta_1 DID_{it} + \gamma Control_{it} + \alpha_t + \lambda_i + \varepsilon_{it} \quad (4)$$

The empirical results corresponding to these mechanisms are shown in Table 6. Column (1)–(2) demonstrates that the BRI has enhanced facility connectivity in participating countries. Facility connectivity focuses on the "hard connection" of infrastructure, which directly improve host countries' electricity generation capacity through energy-related project collaborations. Column (3)–(4) reveals that following the BRI's implementation, both the trade value of PV products and trade facilitation have increased. Unimpeded trade emphasizes market access and regulatory harmonization. The influx of more affordable and high-quality PV products has reduced the barriers to solar power adoption, thereby boosting overall solar power generation. Column (5) illustrates that the BRI has expanded China's outward energy investments. Financial integration highlights "soft connection", addressing host countries'

Table 6
Mechanism test.

	(1)	(2)	(3)	(4)	(5)
	FC_Lag	FC_Grid	UT_Trade	UT_LPI	FI
<i>DID</i>	0.376*** (3.33)	0.073*** (2.64)	0.344*** (2.71)	0.053** (1.99)	0.015** (2.44)
<i>Constant</i>	-21.540*** (-2.87)	6.176** (2.10)	-46.948*** (-5.81)	1.753 (0.95)	-0.509*** (-3.69)
<i>Control</i>	YES	YES	YES	YES	YES
<i>Country FE</i>	YES	YES	YES	YES	YES
<i>Year FE</i>	YES	YES	YES	YES	YES
<i>Observations</i>	2016	762	2300	771	3468
<i>R-squared</i>	0.890	0.720	0.846	0.925	0.276

financing constraints through diversified funding mechanisms, which in turn enhances their production capacity.

4.4. Heterogeneity analysis

4.4.1. Trade barrier

To examine how international competition influences BRI-led energy cooperation, this paper conducts a heterogeneity analysis from two dimensions: Tariff Barriers Score (TBS) and Non-Tariff Barriers Score (NBS). In both cases, higher scores indicate greater trade facilitation and lower levels of trade barriers. The data is sourced from the report of the Fraser Institute. The sample is divided into groups using the median TBS and NBS values from the year immediately before each country joined the BRI. As presented in Fig. 3, the BRI effect is more pronounced in countries with higher TBS. This is because countries with lower tariff barriers typically have more lenient market access conditions, allowing the BRI to effectively reduce residual marginal transaction costs, thereby enhancing policy effectiveness. Similarly, a greater influence in countries with higher NBS is exerted, reflecting lower non-tariff barriers and more developed institutional environments. This can be attributed to the fact that these countries generally possess more developed institutional frameworks, enabling smoother implementation of BRI-driven energy cooperation and generating more substantial positive outcomes.

4.4.2. Resource endowment

This study investigates the heterogeneous effects of resource endowment with respect to two dimensions: solar energy availability and dependence on traditional energy sources. Solar energy availability is measured by annual average solar radiation, derived from data

obtained via Google Earth Engine. To improve efficiency, values from the 5th, 15th, and 25th of each month are used to calculate national annual averages. In particular, variations in solar irradiance entail distinct site selection priorities across different electrification contexts, which may fundamentally influence the impact of the BRI. Therefore, on-grid and off-grid power generation are analyzed separately. Conventional energy dependence is defined as the share of non-renewable electricity generation in total electricity generation, since the share of nuclear and other non-fossil energy sources in the sample is minimal and does not structurally bias the analysis. Data are sourced from the IEA. As illustrated in Fig. 4, the BRI significantly boosts off-grid generation in areas with high solar resources, whereas for on-grid power generation, the BRI consistently demonstrates robust impacts across various regions. This divergence can be attributed to the fact that off-grid systems are mainly used in remote or rural areas and rely heavily on sunlight. As a result, BRI investments are more likely to yield measurable increases in power generation only where solar irradiance is sufficiently high. In contrast, on-grid systems benefit from grid integration and infrastructure flexibility, which reduce their sensitivity to localized resource variability. Furthermore, the BRI has a pronounced effect in areas lacking traditional energy, likely because regions with low dependence on conventional energy exhibit greater receptivity and capacity to expand solar energy.

4.4.3. Regional Heterogeneity

Geographical variation may result in heterogeneous impacts on cooperation intensity and solar energy development under the BRI. Based on the World Bank's regional classification and considering geography and sample size, this study groups countries into five regions: East Asia & Pacific (EAS), Europe & Central Asia (ECS), Sub-Saharan Africa (SSF), Latin America & Caribbean (LCN), and South Asia, Middle East & North Africa (SAS&MEA). As illustrated in Fig. 5, the BRI exerts a significant positive effect on solar power generation in EAS and ECS, but not in the other regions. The EAS region, serving as a core node of the "21st Century Maritime Silk Road", benefits from a well-developed industrial chain, robust energy technology capabilities, and advanced infrastructure, helping BRI projects advance solar deployment. The ECS region—functioning as a key hub of the "Silk Road Economic Belt"—enjoys high levels of cross-border infrastructure connectivity and aligned energy policies, supporting solar collaboration under the BRI. In contrast, other regions face weaker infrastructure and institutional frameworks for solar development, which limits the effectiveness of BRI cooperation.

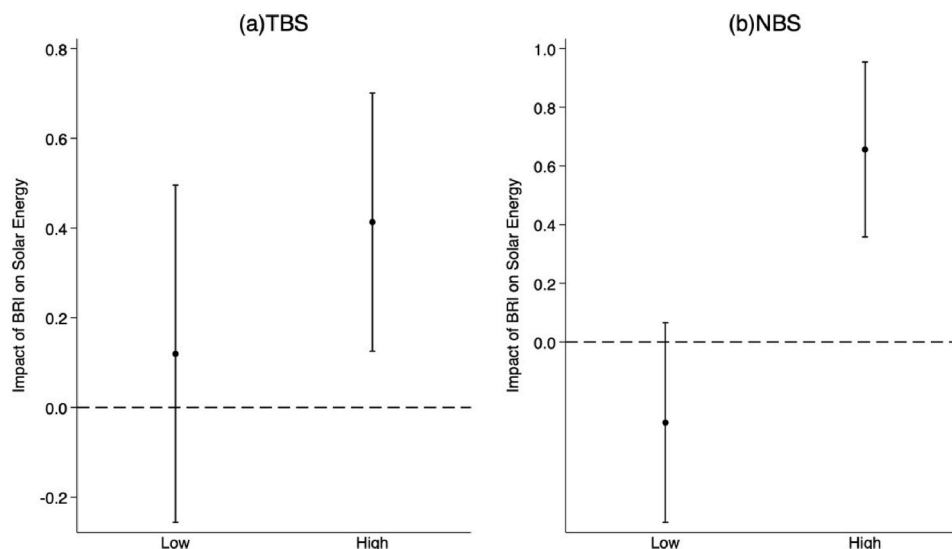


Fig. 3. Trade barrier heterogeneity.

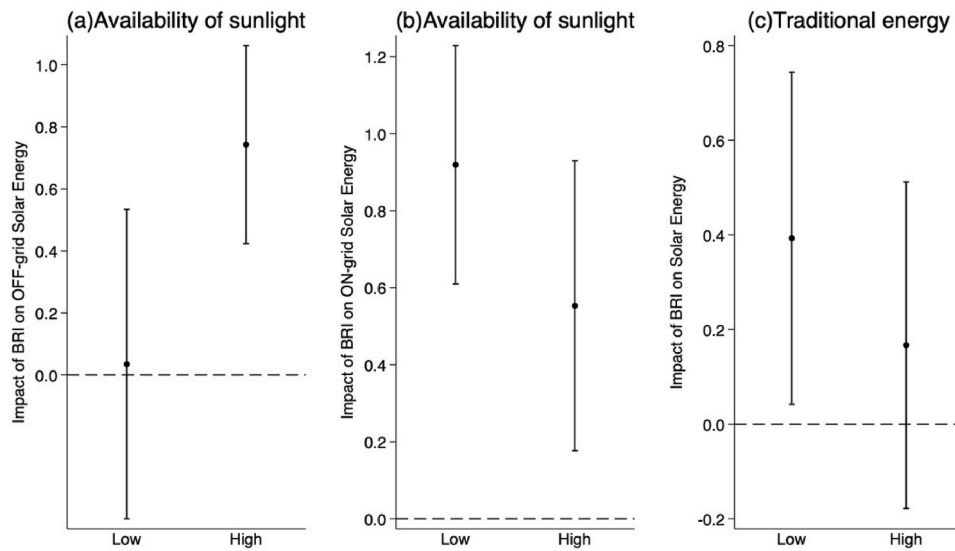


Fig. 4. Resource endowment heterogeneity.

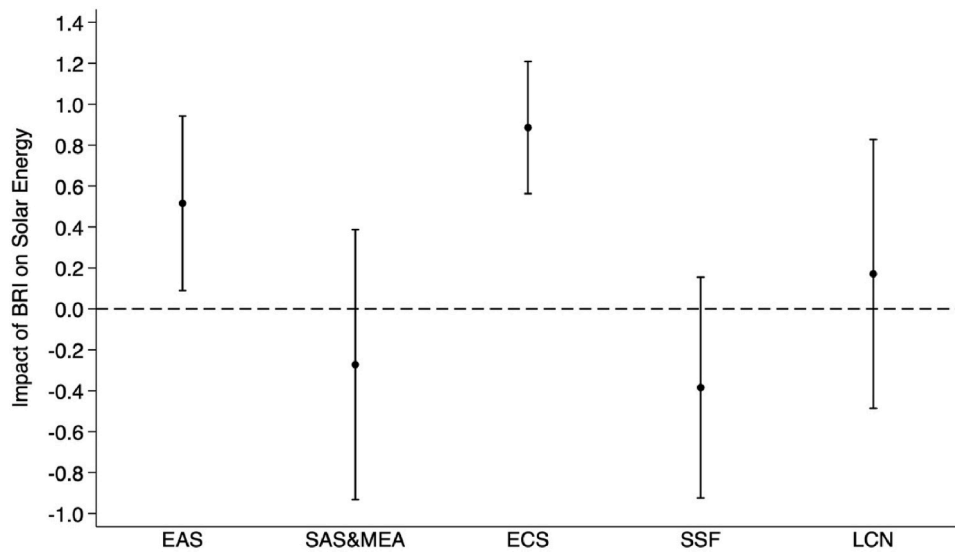


Fig. 5. Regional heterogeneity.

5. Conclusion and policy implications

5.1. Conclusion

The BRI is widely regarded as a key international framework for accelerating the global deployment of solar energy. However, existing research has largely focused on PV trade, with limited systematic evaluation of solar power generation as a tangible outcome of renewable energy cooperation. Using a multi-period DID model, this study rigorously examines BRI's impact on solar power generation, providing empirical evidence of the initiative's tangible contributions and effectiveness. The main findings are as follows:

First, the BRI significantly increases solar power generation in participating countries. This effect remains robust across a range of specifications, including PSM-DID tests. Second, the mechanism analysis reveals that the BRI promotes solar energy development through three main channels: facility connectivity, financial integration and unimpeded trade. Third, the heterogeneity analysis reveals that the BRI exerts a stronger positive impact on solar power generation in countries with

lower trade barriers, while its policy effects are more pronounced in regions characterized by abundant solar irradiance and reduced reliance on traditional energy sources.

5.2. Policy implications

First, given BRI's significant and robust positive impact on boosting solar power generation in participating countries, the Chinese government ought to prioritize BRI-driven solar cooperation. Particularly with recent cost reductions and technological maturation of distributed PV—readily deployable across many low-income BRI nations—China should leverage its solar advantages to consolidate policy gains, advancing BRI's long-term green transition goals. Participating countries must also proactively seize solar development opportunities under this initiative.

Second, this study finds that BRI influences solar power generation through both "hard" and "soft" connectivity mechanisms. Future policymaking should therefore prioritize coordinated advancement and deep integration across these channels. For example: in facility

connectivity, advance joint construction of power grids and energy storage with participating countries to reduce systemic constraints on solar generation; in unimpeded trade, optimize customs procedures and enhance mutual recognition of green certification standards, improving circulation efficiency and brand recognition of Chinese PV products in BRI markets; in financial integration, channel more green investment to critical segments of the solar industry chain, further leveraging multi-lateral development banks and China's green financial system.

Third, regarding the heterogeneous effects, participating countries along the BRI should strengthen their capacity to align with BRI-driven solar energy cooperation by leveraging their distinct institutional and environmental contexts. Concerning trade barriers, for countries with high barriers, green cooperation should be used as an entry point to negotiate reductions through multilateral and bilateral mechanisms, thereby unlocking the full potential of BRI in promoting solar energy development. For countries with low trade barriers, where the BRI effect is more pronounced, governments should proactively expand collaboration in high-value domains such as solar technology and energy storage systems, reinforcing the sustainability gains generated by BRI engagement. Regarding resource endowments, nations should capitalize on their comparative advantages to foster cross-border resource complementarity and facilitate the integration of regional renewable energy supply chains through the BRI platform. The spatial heterogeneity of BRI's green effects, with its positive influence concentrated in core corridors, suggests that efforts should be made to facilitate the diffusion of experiences in solar technology, standards, and project management from East and Central Asia to other regions. At the same time, greater policy and financial support should be directed toward peripheral areas to better address financing constraints and institutional

barriers in regions such as the Middle East, Africa, and Latin America.

5.3. Limitations

While this study provides valuable empirical evidence, several limitations remain. The heterogeneity analysis focuses on broad tariff and non-tariff barriers to reflect cross-country differences in trade institutions. However, solar-related trade frictions are complex, particularly due to specific non-tariff barriers such as technical standards and regulatory requirements. Future research should incorporate these indicators and broaden the understanding of trade institutions.

CRediT authorship contribution statement

Boqiang Lin: Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Conceptualization. **Jinhan Li:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

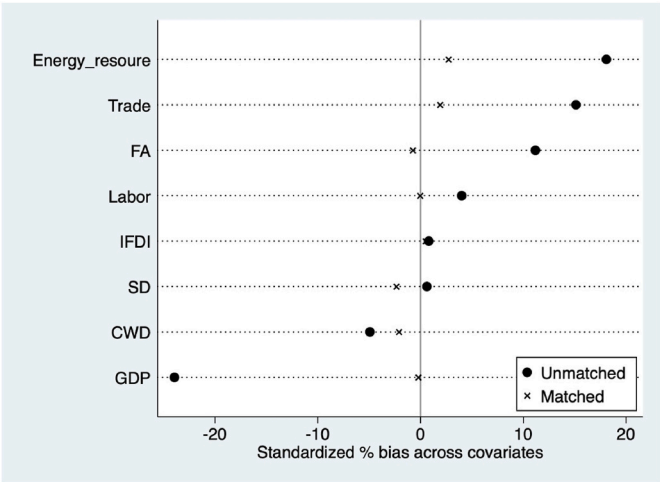


Fig. A.1. Standardized variable deviations before and after kernel matching.

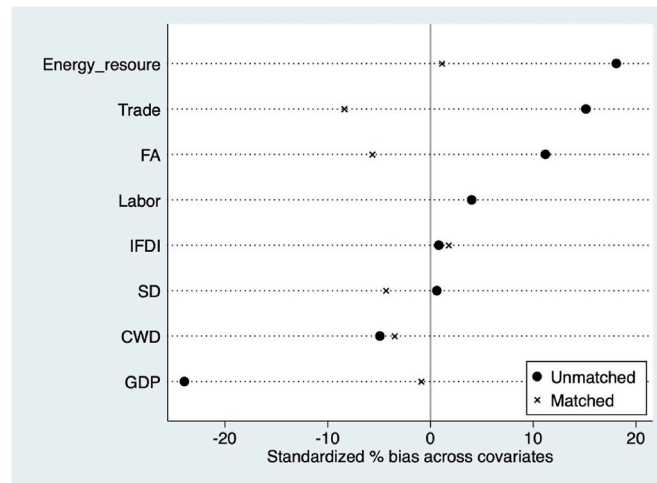


Fig. A.2. Standardized variable deviations before and after nearest neighbor matching.

Data availability

Data will be made available on request.

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